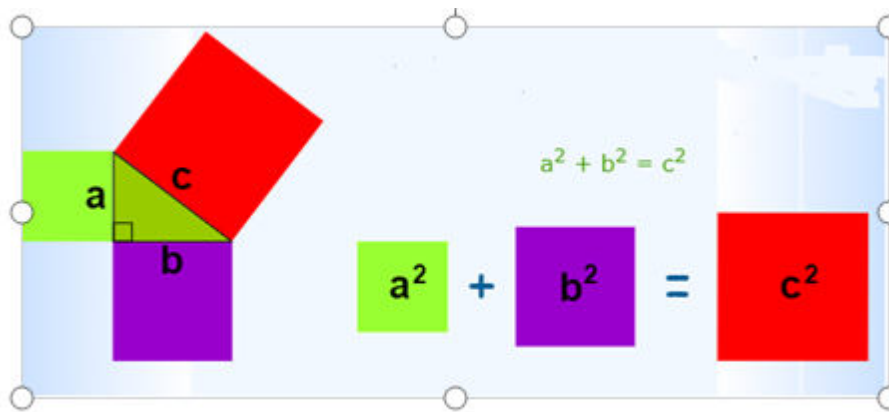
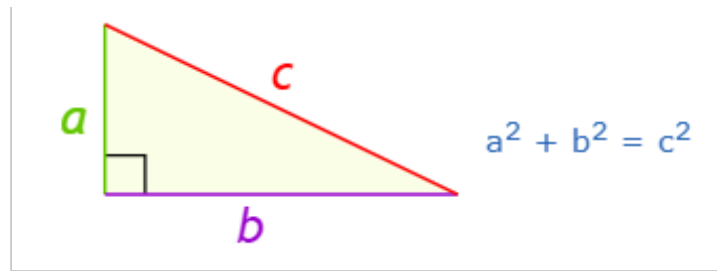
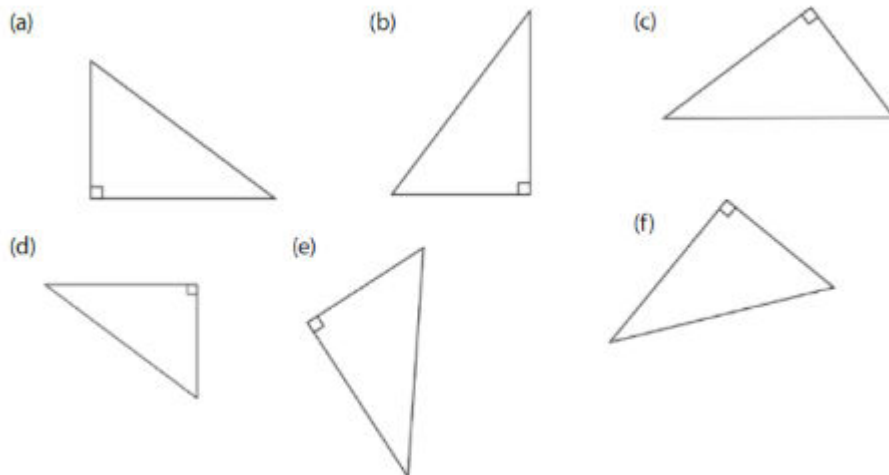


The Pythagoras Theorem states that the square of length of the hypotenuse (c^2) is equal to the sum of the squares of the length of the other two sides (a^2, b^2).



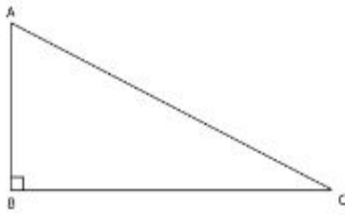
Question 1:

Identify the hypotenuse in the following right angle triangles:



Question 2:

1. Consider the right-angled triangle given below:



2. Measure and write down the length of all the three sides of the right-angled triangle ABC.

AB = _____ cm

BC = _____ cm

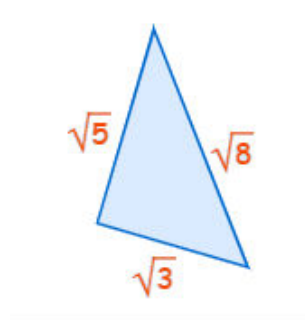
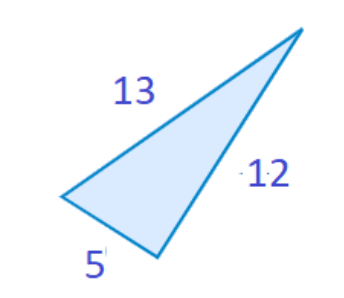
AC = _____ cm

3. Which is the longest side of the right-angled triangle ABC? _____
4. Which side of the right-angled triangle ABC is opposite to the angle of 90° ? _____

Question 3:

Does these triangles have a right angle (using Pythagorean theorem)?

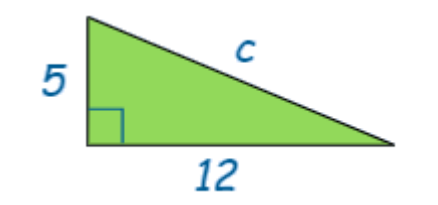
(b)



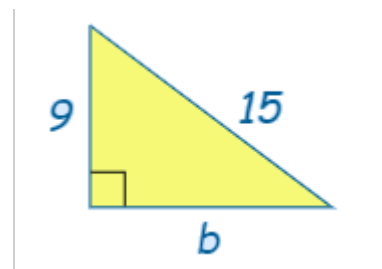
Question 4:

Find the missing sides in the following triangles:

(a)

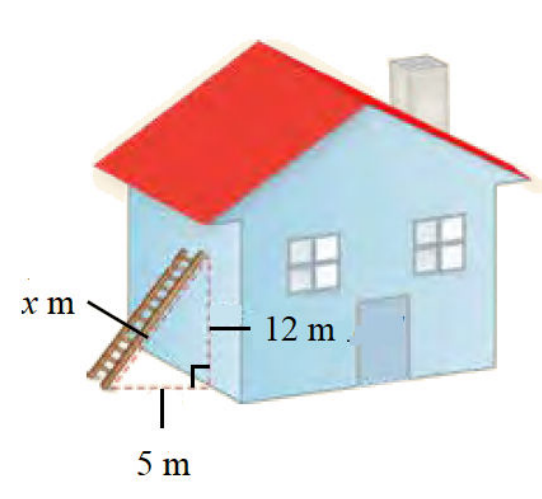


(b)

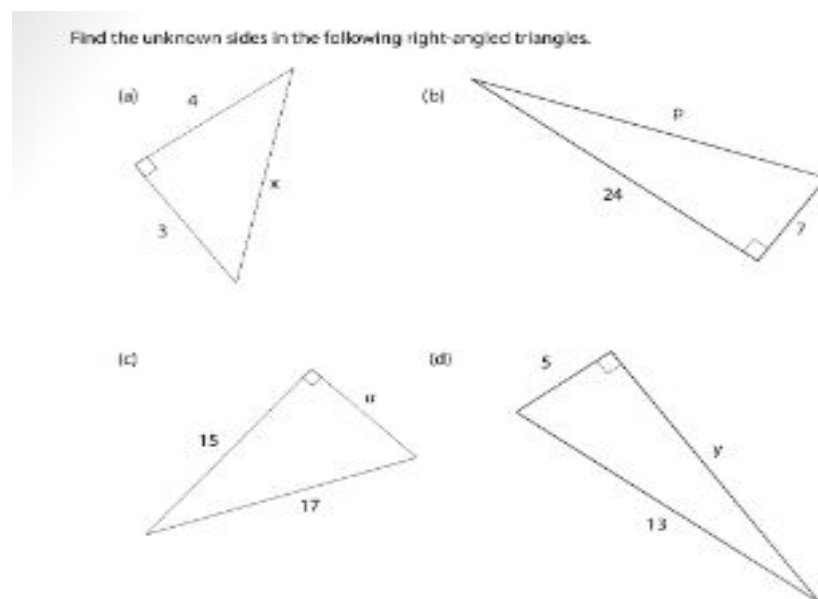


Question 4:

Deborah is trapped in a fire on the third floor of a building, 12 m high. A team of firefighters comes to rescue her. They have 2 ladders of length 13 m and 20 m. Which ladder should they use to reach the 3rd floor if the ladder is placed 5 m away from the building?

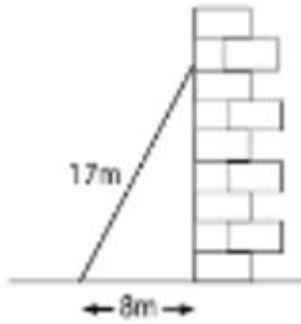


Question 5:



Question 6:

A ladder 17 cm long, leans against a wall. The foot of the ladder is 8 cm from the base of the wall. How far up the wall does the ladder reach?



Question 7:

A ladder 13 m long leans against a wall. The foot of the ladder is 5 m from the base of the wall, How far up the wall does the ladder reach?

